

AMR Clinical Decision Support System

Patient-specific antibiotic resistance prediction — a fast, explainable second opinion for frontline clinicians.

GROUP-6-UQU · Umm Al-Qura University, Dept. of Computer Science · CS6117 Advanced Topics in AI

Ensemble ML (Gradient Boosting) + LLaMA 3.3 70B (Groq) + SHAP · Accepted — Discover AI (Springer Nature, in press)

The Problem

1M+

deaths a year linked to AMR

48–72 hrs

until lab culture results return

Empirical prescribing, under pressure

- Clinicians must pick an antibiotic before culture results come back.
- Pick wrong, and treatment fails — the infection worsens.
- Resistant bacteria spread even faster.

Who It's For



ER Physicians

Triage and admit under time pressure



Hospitalists

Manage inpatients across the wards



ICU Teams

Treat the sickest, sepsis-prone patients

The gap today: existing tools are static, hospital-wide antibiograms — they ignore this patient's labs, comorbidities, and prior antibiotic exposure. Too generic to trust at the bedside.

The Solution

It predicts patient-specific resistance probabilities — then turns them into an **explainable, actionable recommendation in seconds.**



Calibrated ML Ensemble

LightGBM lead — XGBoost,
CatBoost &
HistGradientBoosting

Isotonic-calibrated confidence scores



SHAP Explainability

Reveals which patient features
drove each score

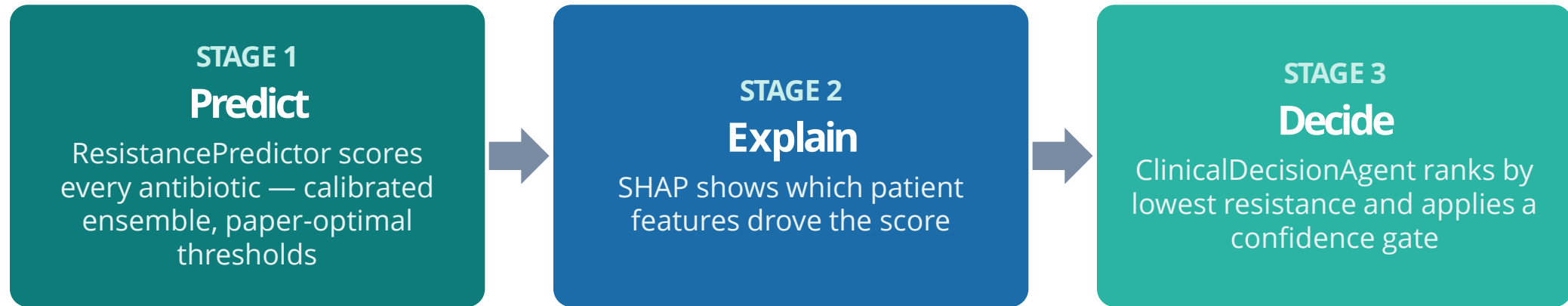


AI Reasoning Agent

LLaMA 3.3 70B (Groq) ranks
options and writes the clinical
rationale

Validated in peer-reviewed research accepted to Discover AI, Springer Nature.

Architecture & Responsible AI



RESPONSIBLE AI

Augments, never overrides

Clinical judgement always stays in control

Explainable & traceable

Every recommendation traces to real patient data

Refuses to guess

No safe option → flags for Infectious Disease review

Built on Peer-Reviewed Research

997
Patients

1.2M+
Observations

20
Organisms

28
Antibiotics

How the models were validated

- Calibrated ensemble — LightGBM lead model; XGBoost, CatBoost & HistGradientBoosting available
- Isotonic calibration turns scores into clinically meaningful confidence levels
- Patient-level cross-validation prevents data leakage across the cohort
- Paper-validated optimal thresholds drive the agent's confidence gate

"Comparative Evaluation of Ensemble ML Models for Predicting Antibacterial Resistance from EHR" · Accepted, Discover AI (Springer Nature, in press) · Almalki, Althaqafi & Al-Shareef

Live Demo — A Real Case

THE CASE

- 65–74-year-old male
- Diabetes (complicated) + chronic kidney disease
- E. coli on culture — Site: Blood (Bacteremia / Sepsis)
- qSOFA 0/3 — Low (updates live); no prior antibiotics

Evaluated: full ~24-drug panel — carbapenems, cephalosporins, fluoroquinolones, aminoglycosides & more

THE RECOMMENDATION

Ertapenem

HIGH CONFIDENCE

0.2%

predicted resistance probability — ranked #1 of the panel

SHAP (LightGBM TreeExplainer) shows the drivers; LLaMA 3.3 writes the rationale.

Caveats: monitor renal function (median_cr 1.0); watch C. difficile risk.

If no antibiotic clears the paper-optimal confidence threshold, the agent refuses to guess and flags the case for Infectious Disease review.

RESEARCH-BACKED · EXPLAINABLE · BEDSIDE-READY

A fast, explainable second opinion in the fight against resistance — one prescription at a time.

WHAT'S NEXT

Live EHR integration

Connect directly to hospital electronic health record feeds

Broader coverage

Expand to more organisms and antibiotic classes

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Thank you · github.com/ZohoorAlmalkiUQU/clinical-decision-agent-amr